Aurion Bank

Customer Churn Analysis Dashboard

10,127 Credit Card Customers - Behavioral, Engagement & Risk Insights

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Data Source: *Bank Churners (Kaggle) - synthetic credit card dataset provided for educational and portfolio use.*
Raw data not redistributed.

Executive Summary

Customer Churn Analysis

This analysis examines churn patterns across 10,127 credit card customers, combining demographic, behavioral, and product usage data. Overall churn is moderate (16.1%), but significant variation exists across segments and behaviors. Churn is driven primarily by engagement intensity and customer experience, with two high-risk groups emerging: low-utilization customers who gradually disengage, and high-utilization customers who face financial stress or friction. Mid-utilization segments show the highest engagement and the lowest churn, representing the bank's most resilient customer base.

Demographic effects are present but secondary. Income extremes (low and very high), specific education-income intersections, and a small Platinum female cohort show elevated churn. Tenure effects are mild but indicate slight fatigue in long-tenured (>49 months) customers. Product tier influences utilization levels but does not guarantee loyalty.

Behavioral signals provide the clearest predictions. Inactivity and repeated contacts (4-5 interactions) sharply increase churn risk. Most attrited customers show declining transactions and spending several months before churn, highlighting an opportunity for early intervention. Overall portfolio exposure is healthy, with a revolving share of 12.5%.

These insights inform a set of targeted retention actions focused on engagement reactivation, proactive support for high-utilization customers, friction reduction, and early behavior-change detection.

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Portfolio Overview

Portfolio health and churn behavior at a glance

FILTERS

Gender: All

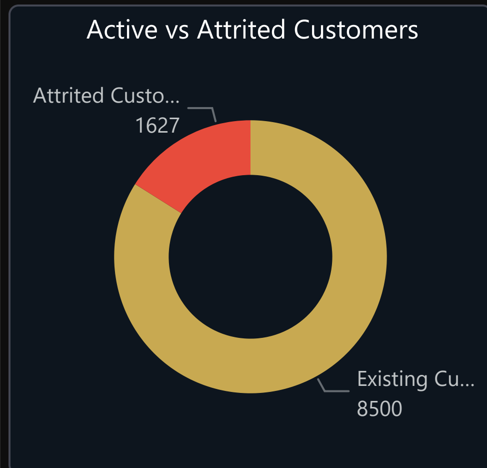
Age: All

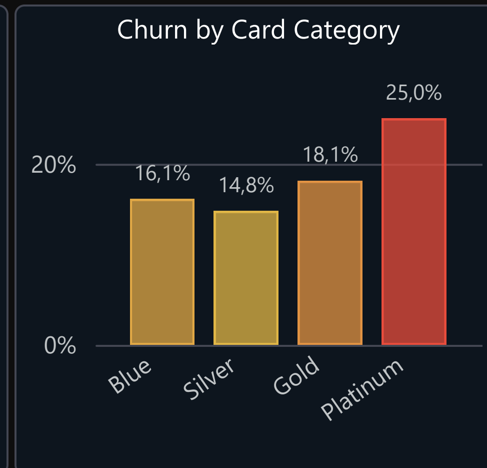
Education: All

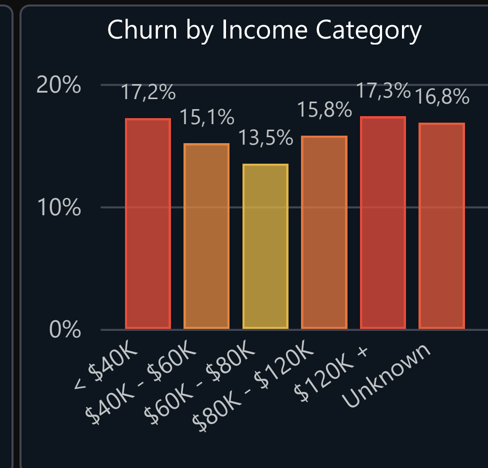
Income: All

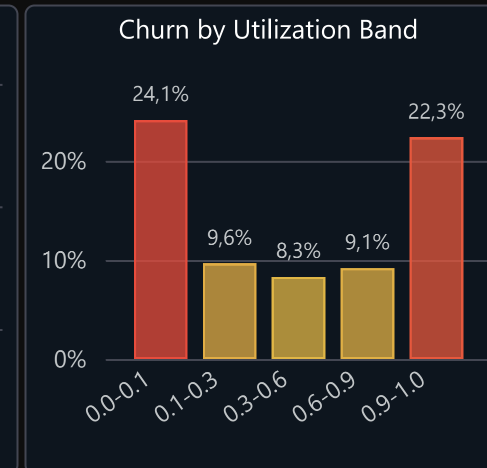
Tenure: All

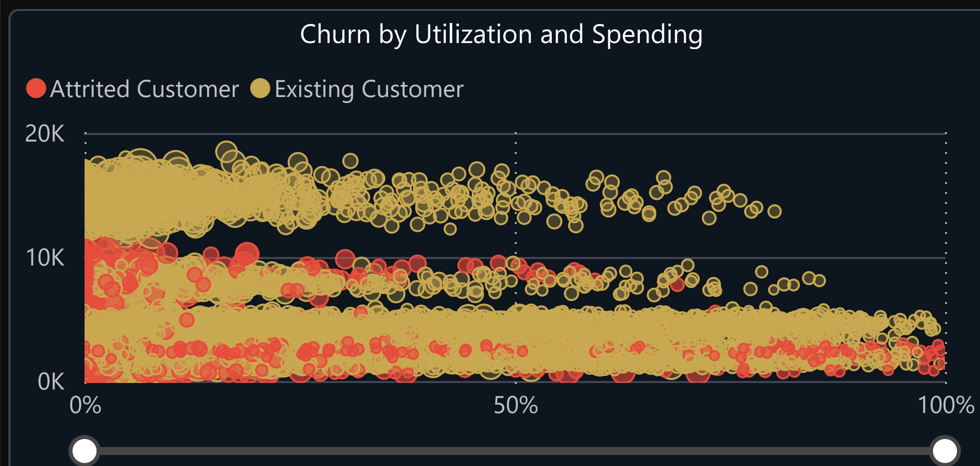
10127	16,1%	36	9,28K	44,6M
Customers	Churn %	Avg Tenure (Months)	Avg Credit Limit	Total Transaction Amount











Insights

1. Churn is moderate (16.1%): Overall retention is stable across the portfolio.
2. Card tier affects risk: Gold and Platinum tiers churn more than Blue and Silver.
3. Income shows a soft churn pattern: Low (<$40K) and very high incomes ($120K+) churn most; middle-income groups are the most stable.
4. Utilization has two risk extremes: Very low (0–10%) and very high (90–100%) utilization show the highest churn.
5. Spending drives retention: The scatter shows that high-spending customers (>11K) almost never churn, while low spenders in extreme utilization bands churn the most.

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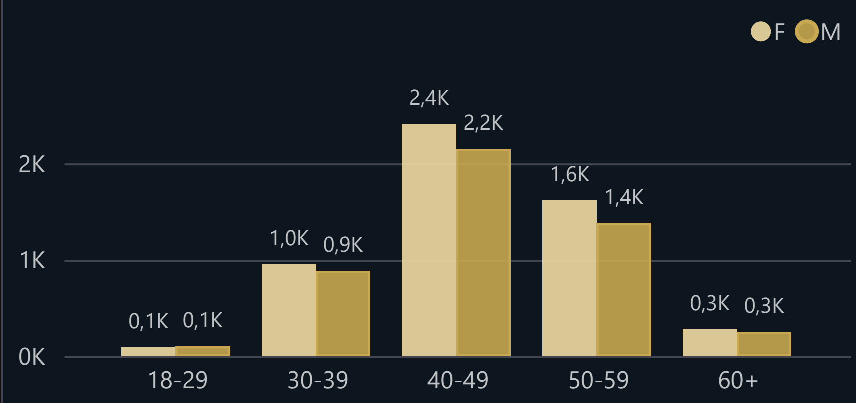
Customer Segmentation

How churn varies across demographics and product tiers

10127	8500	1627	16,1%	25,0%	64,9
Customers	Active Customers	Attrited Customers	Churn %	Avg Utilization %	Avg Transactions / Customer

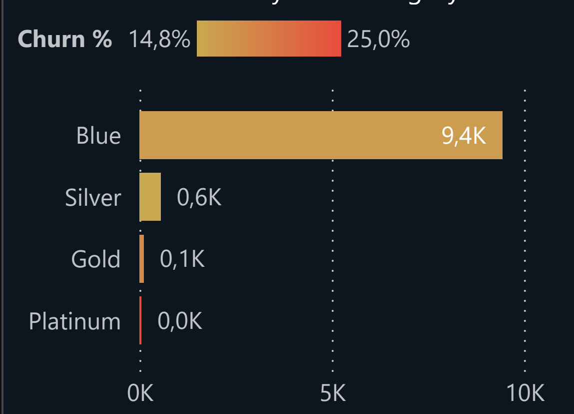
Customers by Age Group and Gender



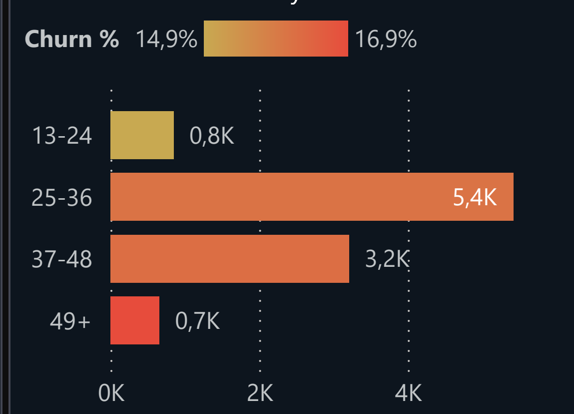
Churn by Education and Income level

Income_Category	College	Doctorate	Graduate	High School	Post-Graduate	Uneducated	Unknown	Total
< $40K	14,2%	26,6%	18,4%	15,1%	20,0%	15,3%	17,3%	**17,2%**
$40K - $60K	13,1%	15,7%	13,9%	14,4%	15,3%	16,5%	18,6%	**15,1%**
$60K - $80K	12,9%	11,9%	12,3%	13,7%	20,8%	12,3%	14,8%	**13,5%**
$80K - $120K	20,6%	19,3%	14,0%	13,3%	21,0%	17,1%	15,1%	**15,8%**
$120K +	17,1%	21,6%	15,7%	19,0%	13,3%	19,3%	15,8%	**17,3%**
Unknown	14,8%	22,9%	14,8%	19,1%	8,5%	17,3%	18,6%	**16,8%**
Total	**15,2%**	**21,1%**	**15,6%**	**15,2%**	**17,8%**	**15,9%**	**16,9%**	**16,1%**

Customers by Card Category



Customers by Tenure Band



Insights

1. Portfolio skews older: Most customers are 40-49, followed by 50-59. Younger segments (18-29) form a small minority.
2. Education × income effects vary sharply: Churn peaks among doctorate holders at extreme income levels (low <$40K or high $120K+), while the 60–80K income group shows the lowest churn overall.
3. Gender differences are modest but consistent: Women churn slightly more than men across most age groups (~3% higher), with the exception of the 60+ segment where female churn is slightly lower. The Platinum female segment shows the sharpest spike, though sample size is small.
4. Card tier influences churn: Blue and Silver customers show moderate churn (16.1% and 14.8%), while Gold rises to 18%. Platinum shows 25% churn, though sample size is very small.
5. Tenure effects remain mild: 1-2 year customers churn the least (14.9%), while very long-tenure customers (49+) show a slight increase (16.9%), indicating mild fatigue.

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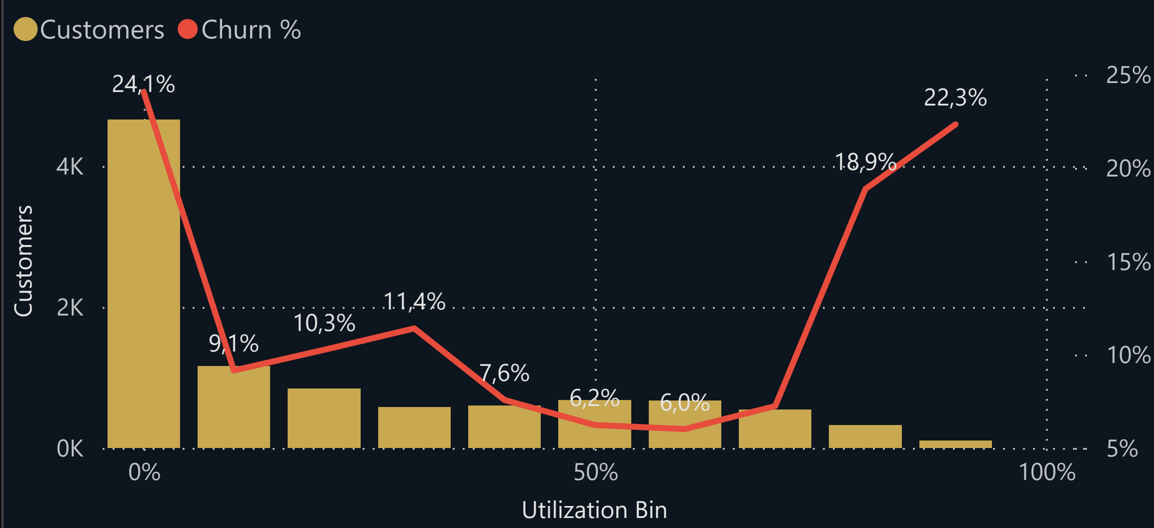
Engagement & Value

How card usage patterns relate to customer loyalty

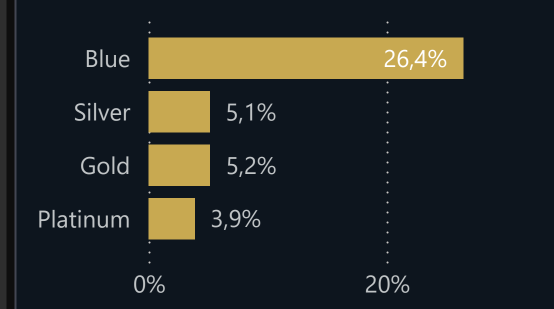
Customers per Utilization Bin with Churn Line



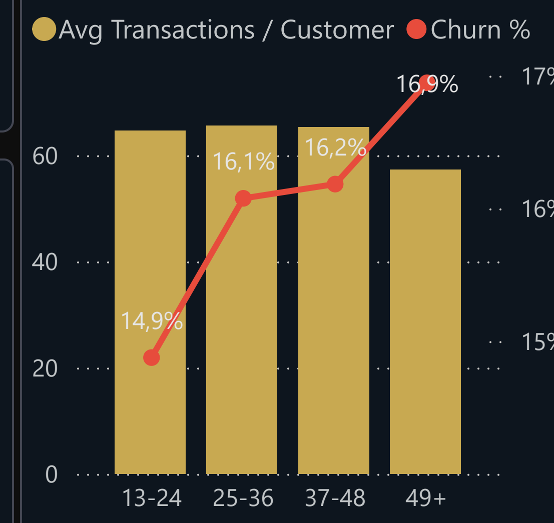
Revolving Share

12,5%

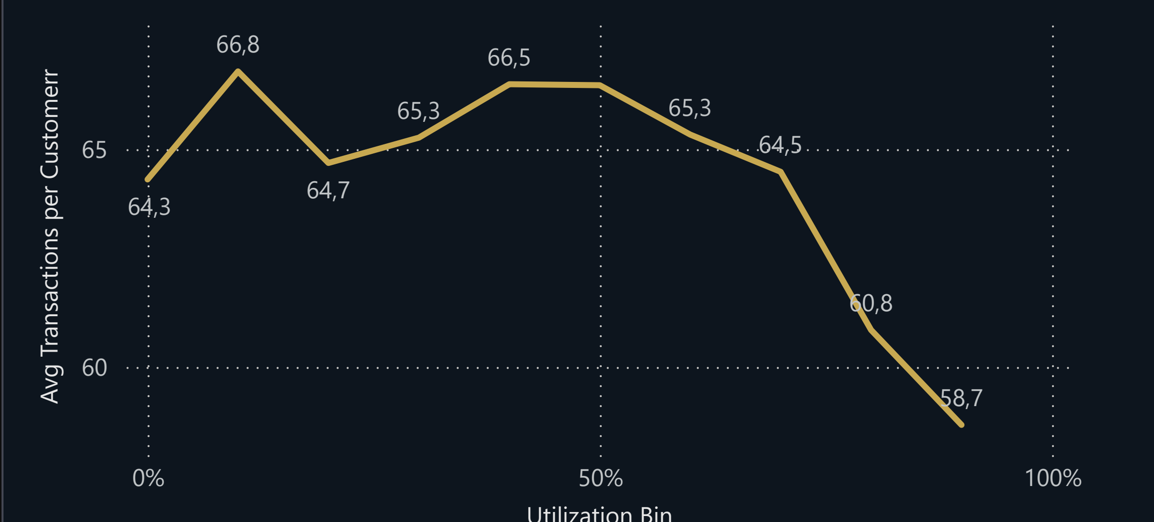
Avg Utilization by Card Category



Avg Transactions by Tenure Band



Avg Transactions by Utilization Bin



Insights

1. U-shaped risk across utilization: Customers with very low (0-10%) and very high (90-100%) utilization show the highest churn rates (~24% and ~22% respectively). Mid-utilization customers (40-70%) have the lowest churn (below 10%).
2. Engagement and churn move in opposite directions: Customers in the mid-utilization range transact the most, while both low-use and maxed-out users show reduced activity and higher churn.
3. Healthy Revolving Share: Customers use 12.5% of total credit limits - a balanced, low-risk portfolio exposure.
4. Blue tier drives utilization: Blue tier customers use 26.4% of their limits, while higher-tier cards stay low (4-5%). Product tier strongly affects usage (Gold and Platinum groups are small and should be interpreted cautiously).
5. Tenure has a mild churn effect: Customers in their first 1-2 years churn the least, while long-tenured customers (49+) show a slight rise in churn (fatigue).

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Drivers & Risk

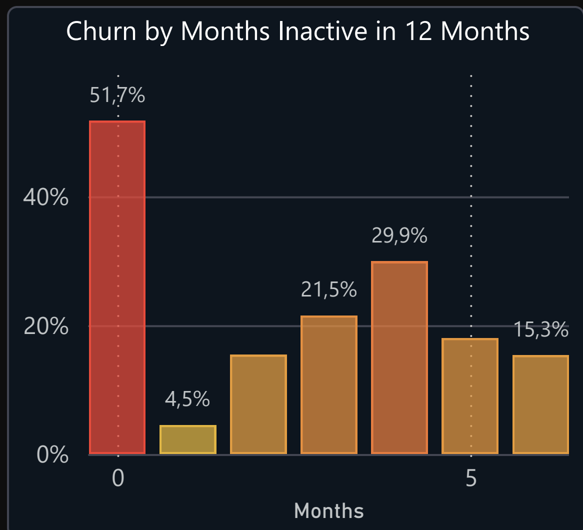
Key behavioral signals that predict churn early

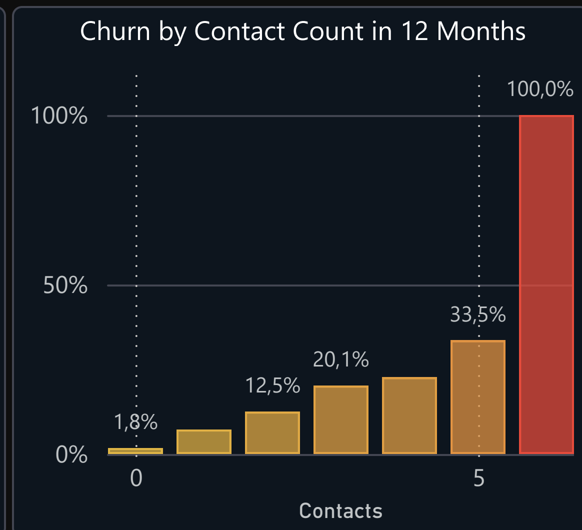
Utilization Threshold Selection

90%

103
High Utilization Customers

1,0%
High Utilization %

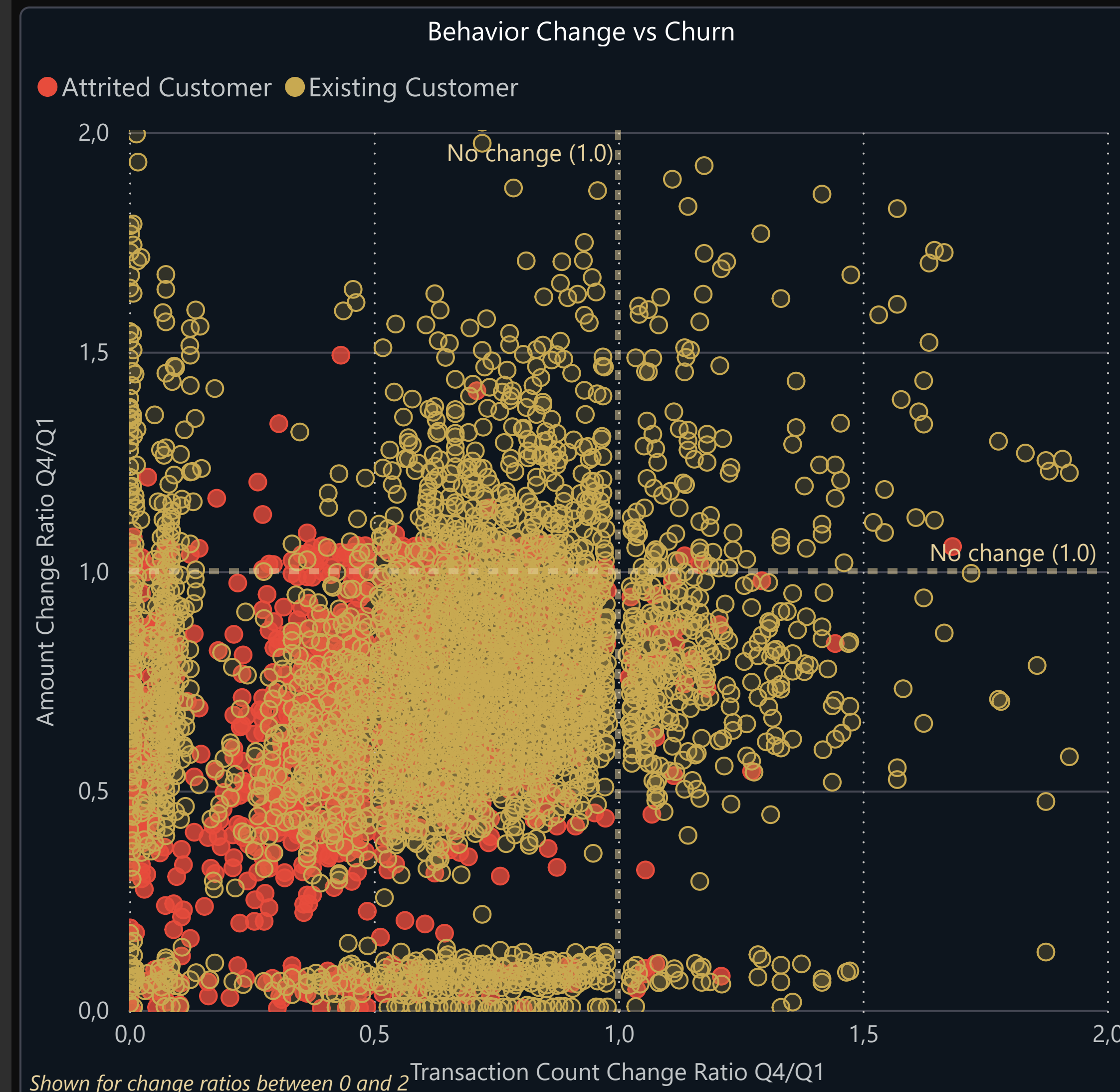




Insights

1. Inactivity strongly predicts churn: Churn climbs rapidly as customers stop using the card - from only 4.5% at 1 month inactive to 29.9% at 4 months inactive (0, 5 and 6-month groups are small and should be interpreted cautiously).
2. Contact frequency is a clear churn escalator: Churn rises almost linearly from 1.8% (zero contacts) to 33.5% (five contacts). The 6-contact spike indicates a tiny, highly escalated segment.
3. Behavior change reveals soft churn patterns: Attrited customers cluster where both transaction count and spending declined vs Q1 (ratios <1.0), indicating gradual disengagement rather than sudden drops.
4. High utilization threshold reveals risk concentration: Adjusting the threshold identifies the share of customers who rely heavily on their card - a key segment for credit and retention monitoring.

Select all | Attrited Customer | Existing Customer



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Recommendations & Next Actions

Targeted strategies to reduce churn and strengthen customer engagement

1. Re-engage low-utilization customers (0-10% range)

- Launch targeted campaigns: small cashback, bill-payment rewards, or activation bonuses.
- Trigger push/email reminders after 1 month of inactivity.
- Offer "first-use" incentives for dormant cards.

2. Support high-utilization customers (90-100% range)

- Proactively review credit limits to reduce friction.
- Provide tailored financial guidance or repayment options.
- Prioritize them for premium support because they are high-value but at higher churn risk.

3. Resolve issues fast for multi-contact customers (4-5 contacts)

- Route repeat callers to specialized agents.
- Introduce "first-call resolution" KPIs.
- Identify common root causes → reduce friction points.

4. Monitor behavioral decline early (Q4/Q1 drop indicators)

- Auto-flag customers whose Tx count & spend ratios fall below 1.0.
- Add an "Early Disengagement Score" combining:
- ↓ Transactions
- ↓ Spending
- ↑ Inactivity
- This can detect churn 1–2 months before it happens.

5. Strengthen loyalty in the mid-utilization segment (40-70%)

- This is your most profitable, lowest-risk group.
- Maintain engagement via habit-building rewards (transport, groceries, utilities).
- Offer tier progression incentives to avoid them slipping into the low-use zone.

6. Tailor retention actions to key demographic groups

- Doctorate + low income: likely financial stress → offer relief options.
- Doctorate + high income: likely value seekers → premium offers.
- Female Platinum users: monitor closely (high churn spike).
- 4+ year tenured customers: mild fatigue → refresh benefits.

7. Focus product improvements on Blue segment

- They represent the majority of cardholders and drive 26,4% utilization.
- Improving their satisfaction has the biggest portfolio-level impact.